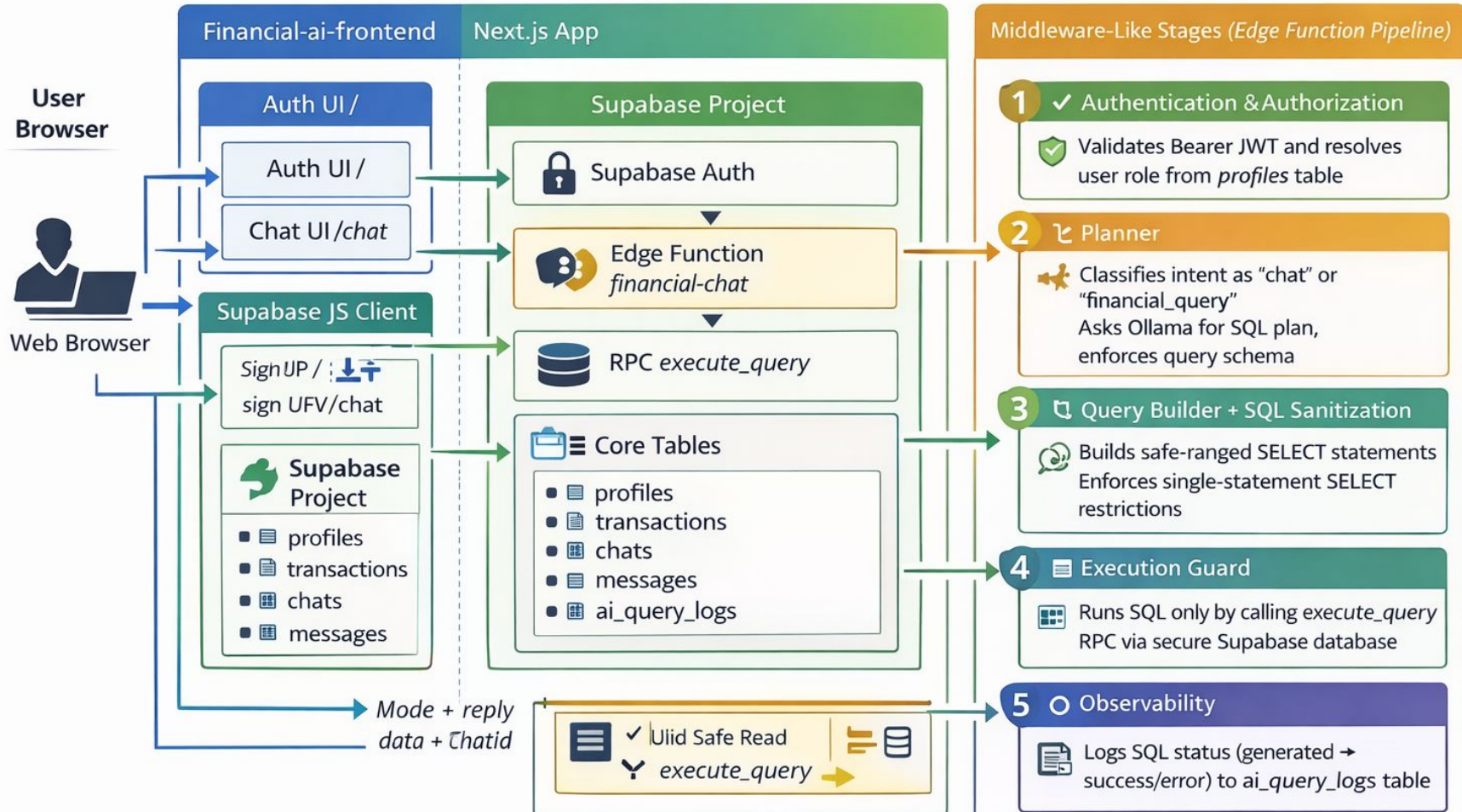


Financial AI System Architecture

Secure, Controlled & Auditable AI for
Financial Insights

Financial AI System - Architecture



High-Level Overview

- AI-powered chat interface for financial insights
- Built with security-first architecture
- AI never directly accesses the database
- Designed for scalability and compliance

User & Frontend Layer

- Users interact via a web browser
- Next.js frontend provides authentication and AI chat UI
- All requests are authenticated using Supabase Auth
- Frontend never directly accesses financial data

Backend & AI Orchestration

- Supabase Edge Function acts as secure AI gateway
- Validates user identity and role
- Controls AI behavior and data access
- Orchestrates AI requests and responses

AI Safety & Query Pipeline

- AI first classifies intent (chat vs financial query)
- Strict SQL sanitization: SELECT-only queries
- Execution only via secure database RPC
- All AI-generated queries are logged

Business Value

- Enterprise-grade security for financial data
- Full audit trail of AI activity
- Accurate and explainable financial insights
- Ready for real-world production use